

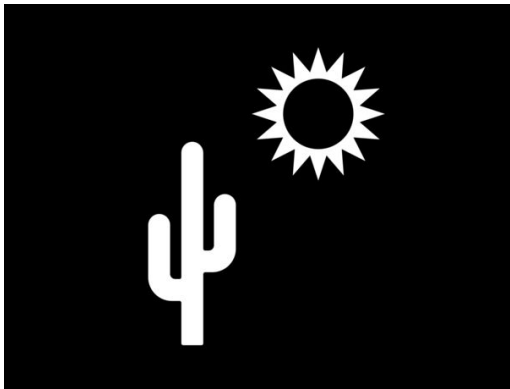
A particle method for solving Fredholm equation of the first kind

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Janvier 2026

General presentation

- inverse problems (Fredholm equation) : $h(y) = \int_{\mathbb{X}} f(x)g(y | x)dx \quad \forall y \in \mathbb{Y}.$
-
- EM(S) algorithm $f_{n+1}(x) = \int_{\mathbb{X}} K(x', x)f_n(x') \int_{\mathbb{Y}} \frac{g(y | x')h(y)}{\int_{\mathbb{X}} f_n(z)g(y | z)dz} dy dx'$
-
- Feynman-Kac formalism

$f(x)$



$g(y|x)$



$h(y)$

Feynman-Kac formalism of EMS

$$M_{n+1}((x_{n+1}, y_{n+1}) | (x_n, y_n)) = K(x_n, x_{n+1})h(y_{n+1})$$

$$G_n(x_n, y_n) = \frac{g(y_n | x_n)}{\int_{\mathbb{X}} \eta_n|_{\mathbb{X}}(z)g(y_n | z) \, dz}$$

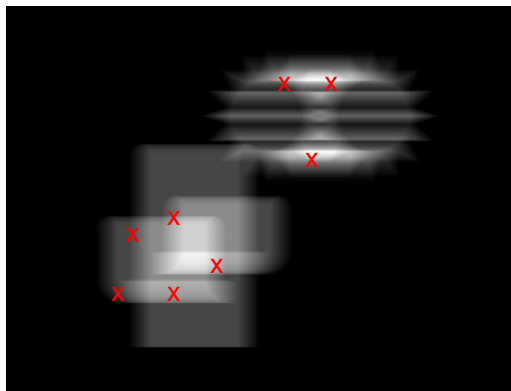
$$\eta_{n+1}(x, y) = \int_{\mathbb{X} \times \mathbb{Y}} \eta_n(x, y) M_{n+1}((x, y) | (x', y')) G(x', y') dx' dy'$$

By augmenting the state space to “sharp” and “blurred” image, we obtain a Feynman-Kac recursion

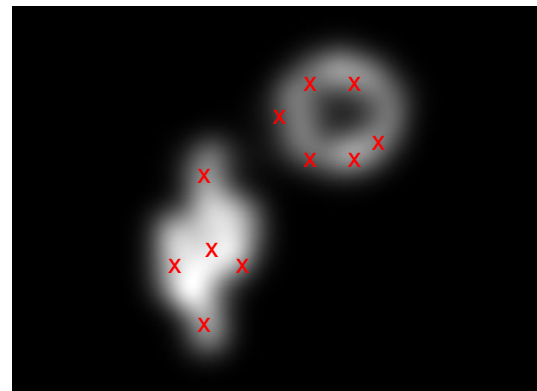
The particle approximation

$$(X_n^i, Y_n^i)_{1 \leq i \leq N}$$

$$\eta_n \rightarrow \eta_n^N = \frac{1}{N} \sum_{i=1}^N \delta_{(X_n^i, Y_n^i)}$$



$$(Y_n^i)_{1 \leq i \leq N} \sim h$$



$$(X_n^i)_{1 \leq i \leq N} \sim f_n$$

Approximation for potentials

- Particles $(X_n^i, Y_n^i)_{1 \leq i \leq N}$ $\eta_n \rightarrow \eta_n^N = \frac{1}{N} \sum_{i=1}^N \delta_{(X_n^i, Y_n^i)}$ $(f_n \rightarrow f_n^N := \frac{1}{N} \sum_{i=1}^N \delta_{X_n^i}, h \rightarrow h^N := \frac{1}{N} \sum_{i=1}^N \delta_{Y_n^i})$
- Approximate potentials

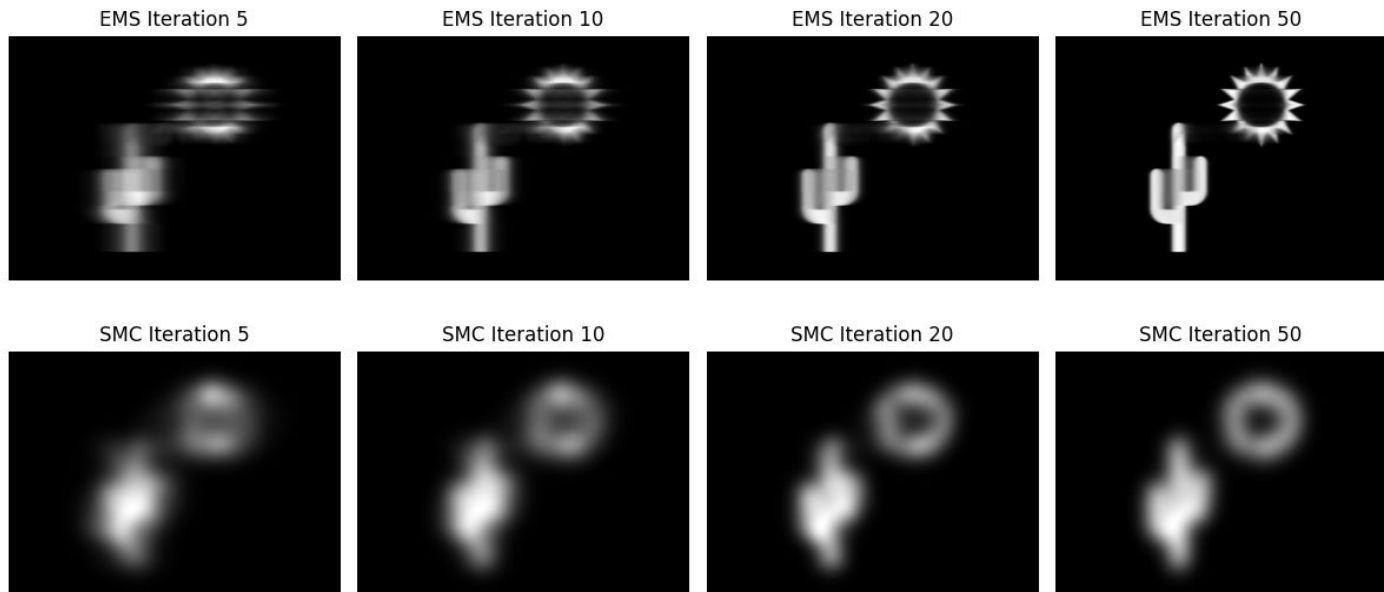
$$G_n(x_n, y_n) = \frac{g(y_n|x_n)}{\int_{\mathbb{X}} f_n(z)g(y_n|z)dz} \rightarrow G_n^N(x_n, y_n) = \frac{g(y_n|x_n)}{\frac{1}{N} \sum_{i=1}^N g(y_n|X_n^i)}$$

Second integral approximation

The quantity $\int_{\mathbb{Y}} \frac{g(y|X_n^i)}{\frac{1}{N} \sum_{k=1}^N g(y|X_n^k)} h(y) dy$

is approximated with Monte-Carlo by sampling a number M of elements Y with density h $(Y_n^i)_{1 \leq i \leq N} \sim h$

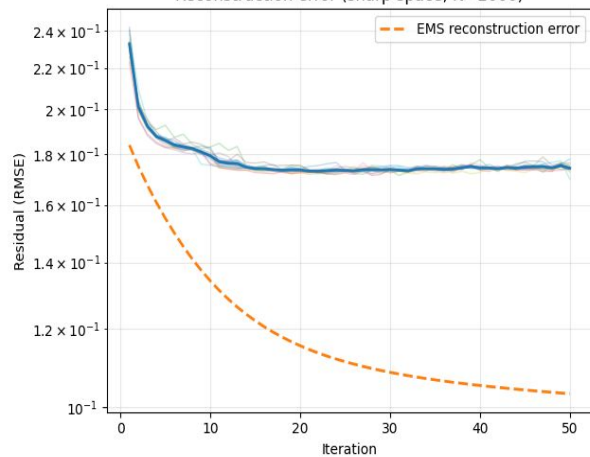
First results



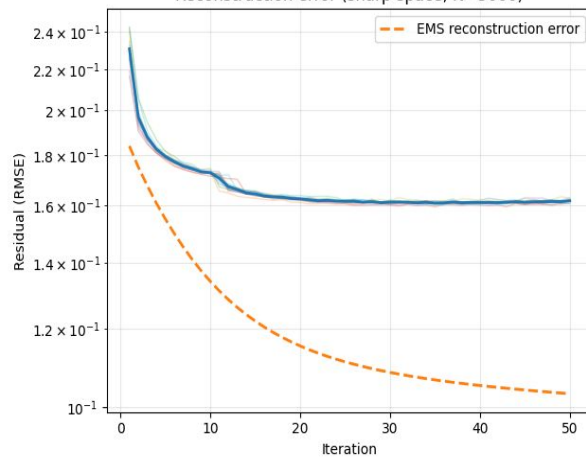
quick fix-point convergence

First results

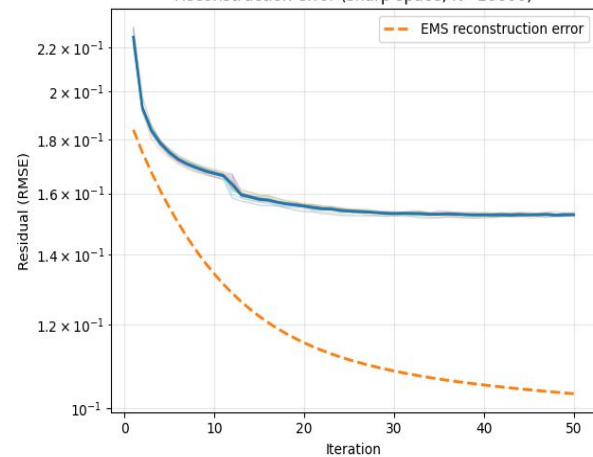
Reconstruction error (sharp space, N=2000)



Reconstruction error (sharp space, N=5000)

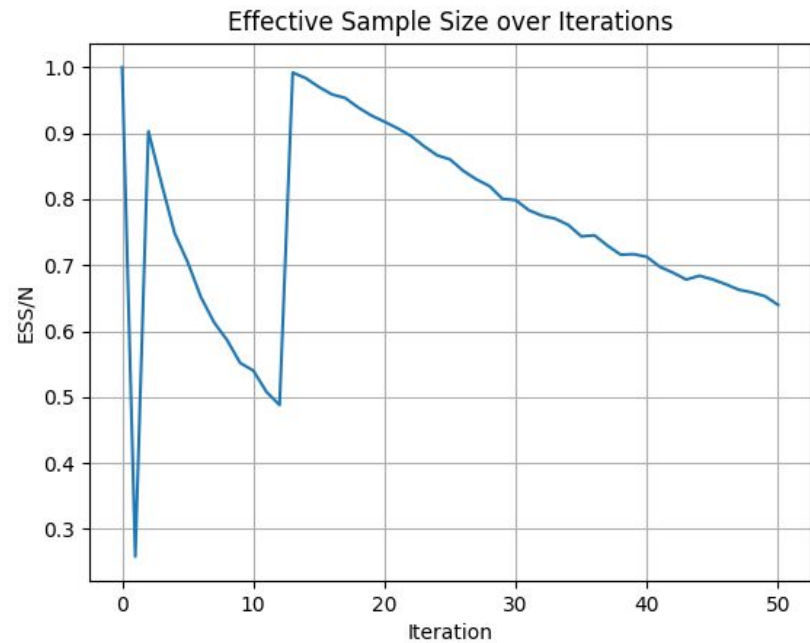


Reconstruction error (sharp space, N=10000)



ESS

multinomial resampling if $ESS < N/2$

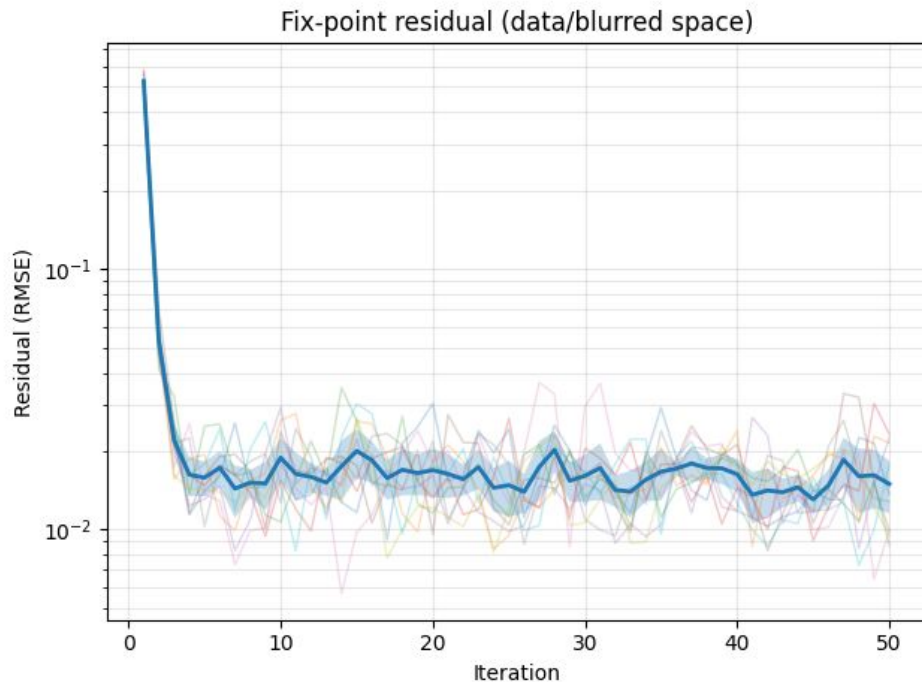


Fix-point convergence

Proposition 1: EMS has a fix-point f^*

We plot $\text{RMSE}(\hat{h}_{n+1}^N - \hat{h}_n^N)$

Plateaus quickly (not to 0 because of Monte-Carlo approximation)



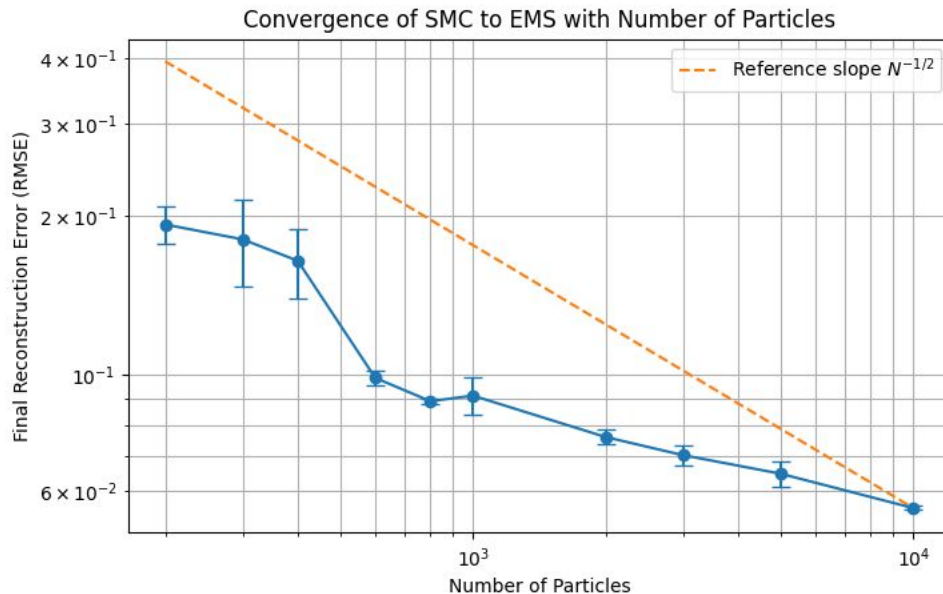
Convergence of SMC towards EMS

Proposition 3 states that

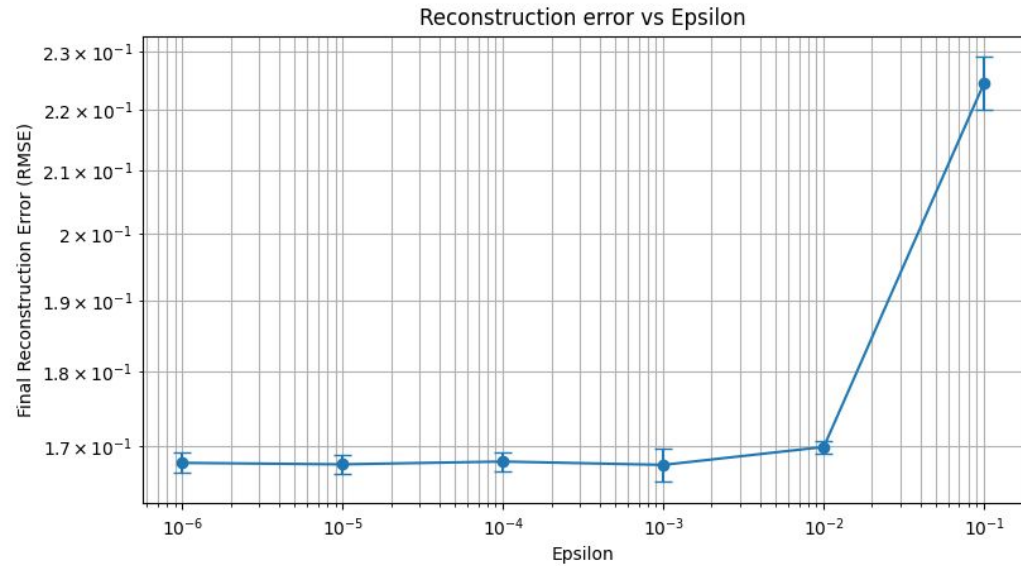
$$\mathbb{E} \left[|\hat{h}_n^N(y) - \hat{h}_n(y)| \right] \leq C_n \frac{\|g(y|x)\|_\infty}{\sqrt{N}}$$

and therefore

$$\text{RMSE}(\hat{h}_n^N - \hat{h}_n) = O\left(\frac{1}{\sqrt{N}}\right)$$



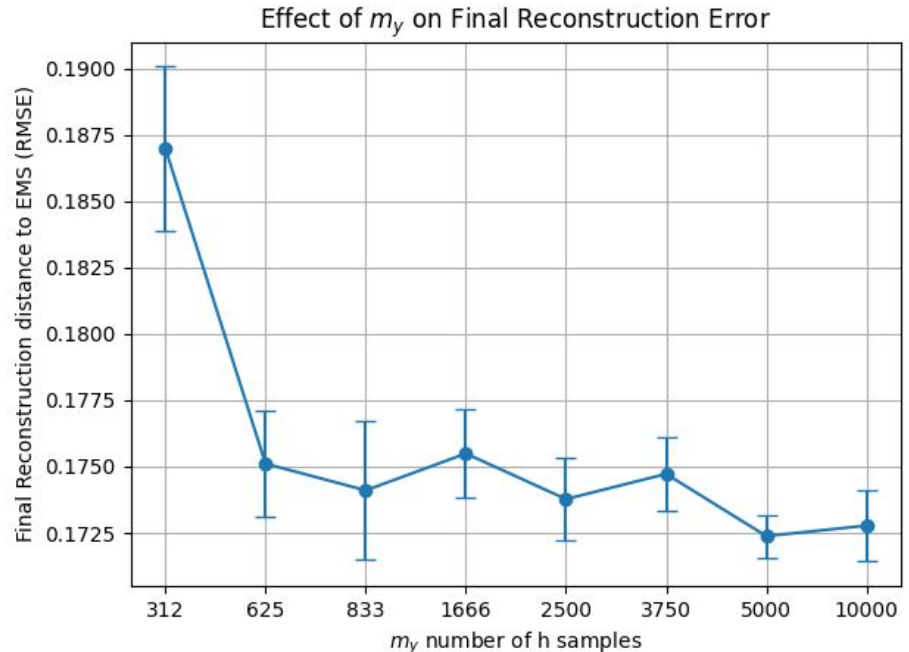
Influence of epsilon



Influence of integral approximation

What if we use less than N particles to approximate the integral of G_n against h ?

What if we use more...

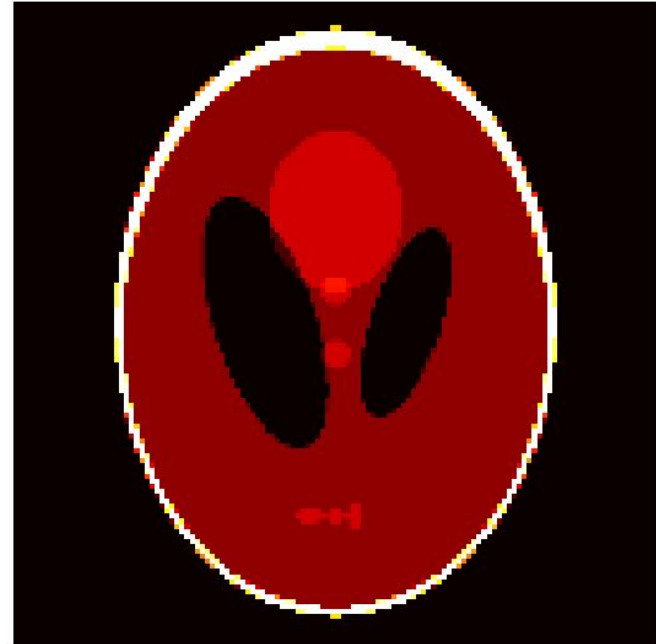


PET Reconstruction via SMC

Reconstruct an image from noisy Radon data

$$h(\phi, \xi) = \int_{\mathbb{X}} \mathcal{N}(x \cos(\phi) + y \sin(\phi) - \xi; 0, \sigma^2) f(x, y) dx dy$$

Phantom



Radon Data & Poisson Noise

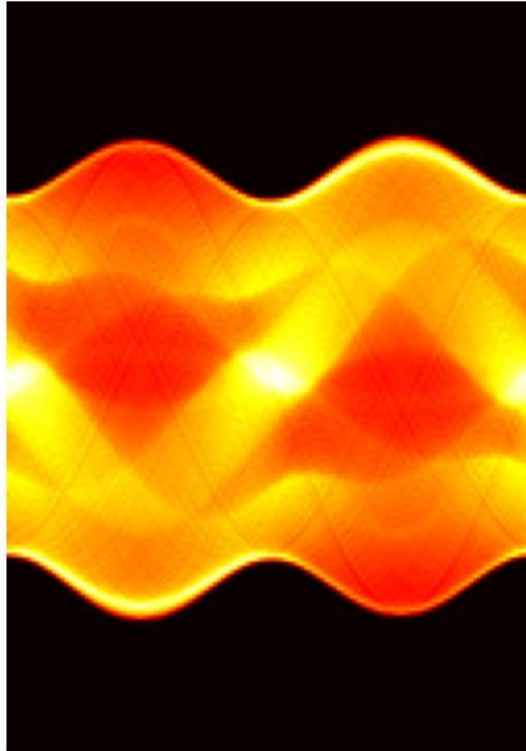
Resolution : $128 * 128$

Number of particles : 20 000

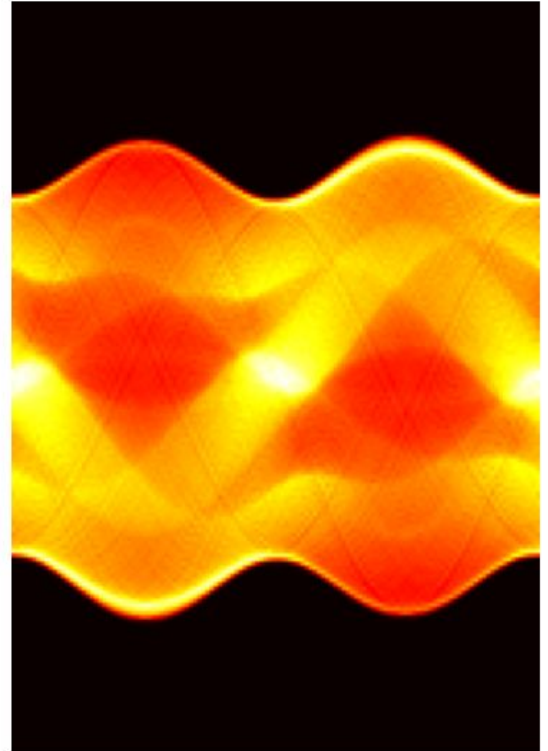
Poisson scale : $5e4$

Sigma : 0.02

Radon data (clean)

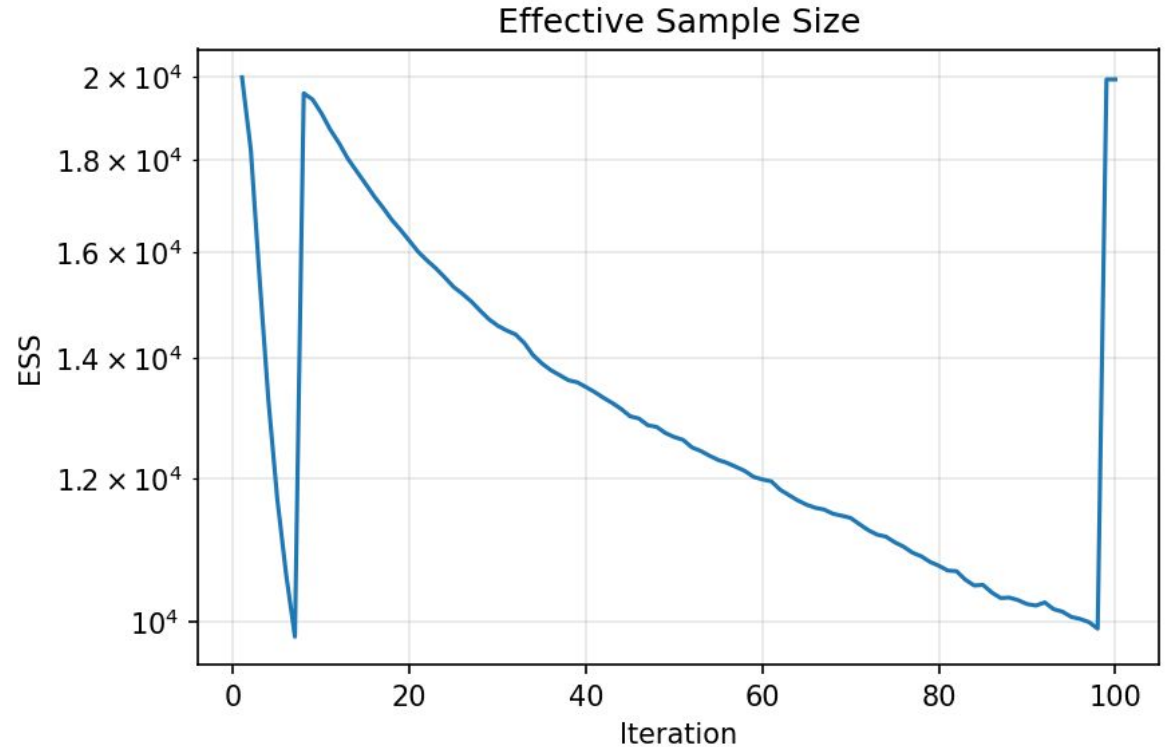


Radon data (noisy)

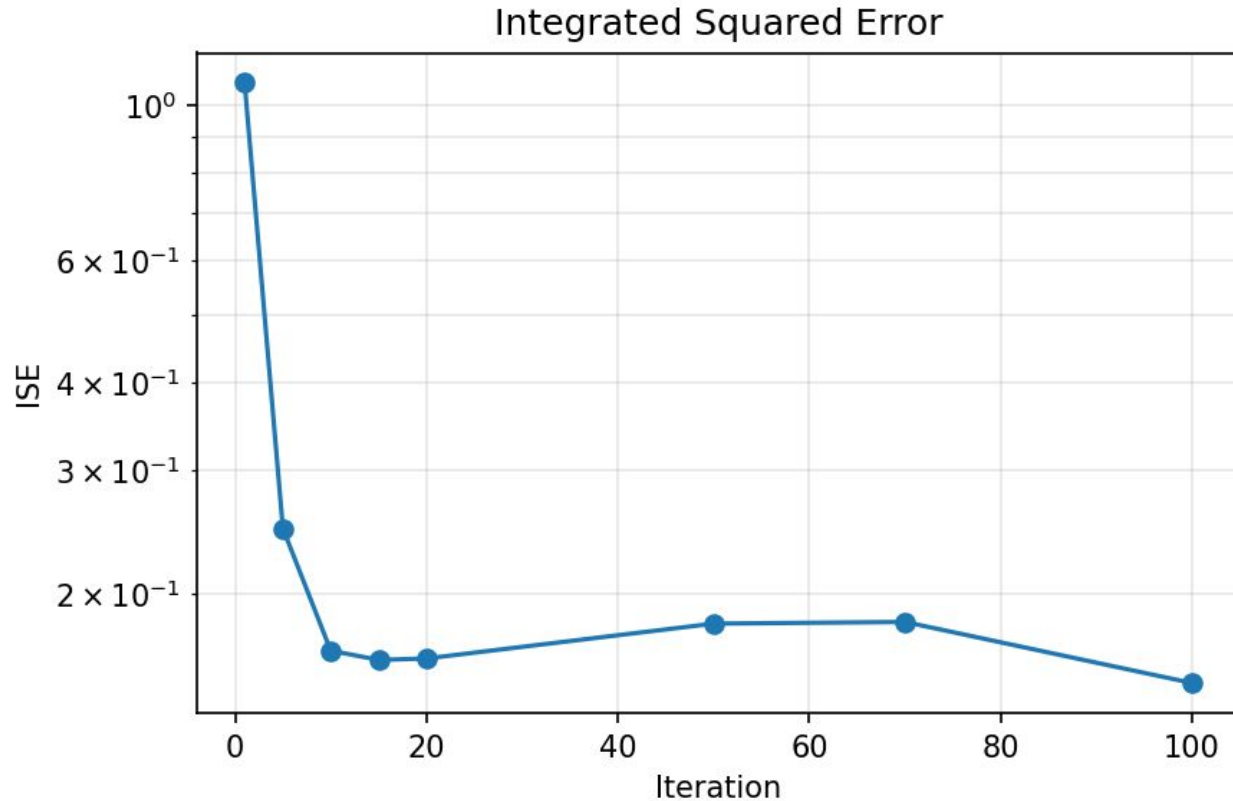


SMC algorithm : ESS curve

Multinomial Resampling
when $ESS < N/2$

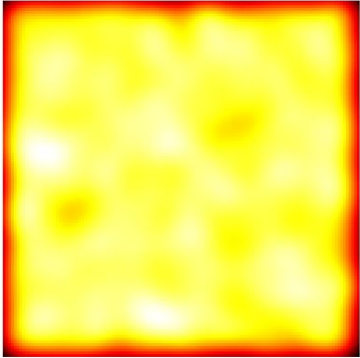


SMC algorithm : Reconstruction Quality

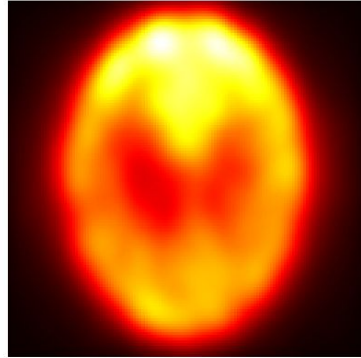


SMC algorithm : Final Results

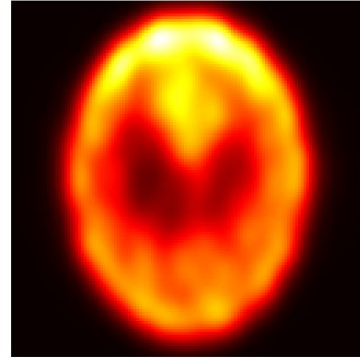
Iteration 1



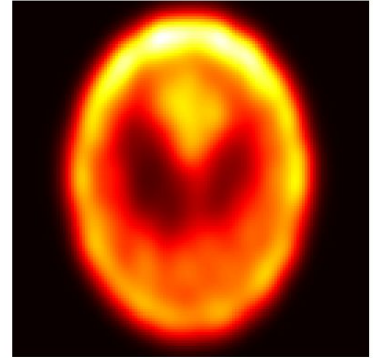
Iteration 5



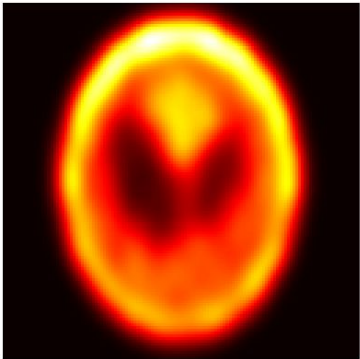
Iteration 10



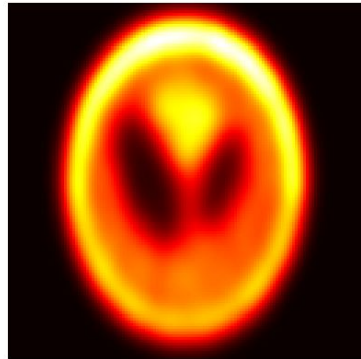
Iteration 15



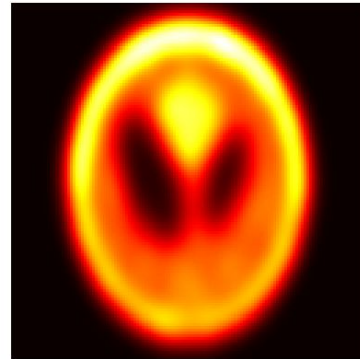
Iteration 20



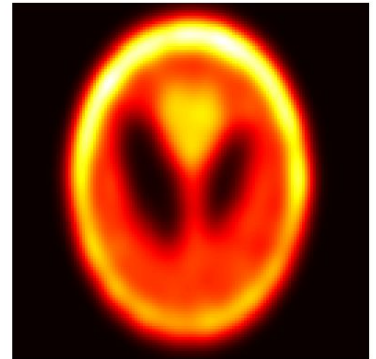
Iteration 50



Iteration 70



Iteration 100



SMC algorithm : Final Results

Computational time : 1 hour for 100 iterations and 20 000 particles

Final ESS : 19 945

Final ISE : 0.149